Healthcare Data Analysis Dashboard

Objective:

Analyze patient data to understand disease trends, hospital performance, and billing patterns.

The objective of this project is to analyze healthcare patient data to identify trends in:

- Patient demographics
- Disease distribution
- Hospital performance
- Billing patterns

The goal is to derive meaningful insights that can support better decision-making in healthcare management.

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Dataset Description:

The dataset contains patient-level healthcare records with the following key attributes:

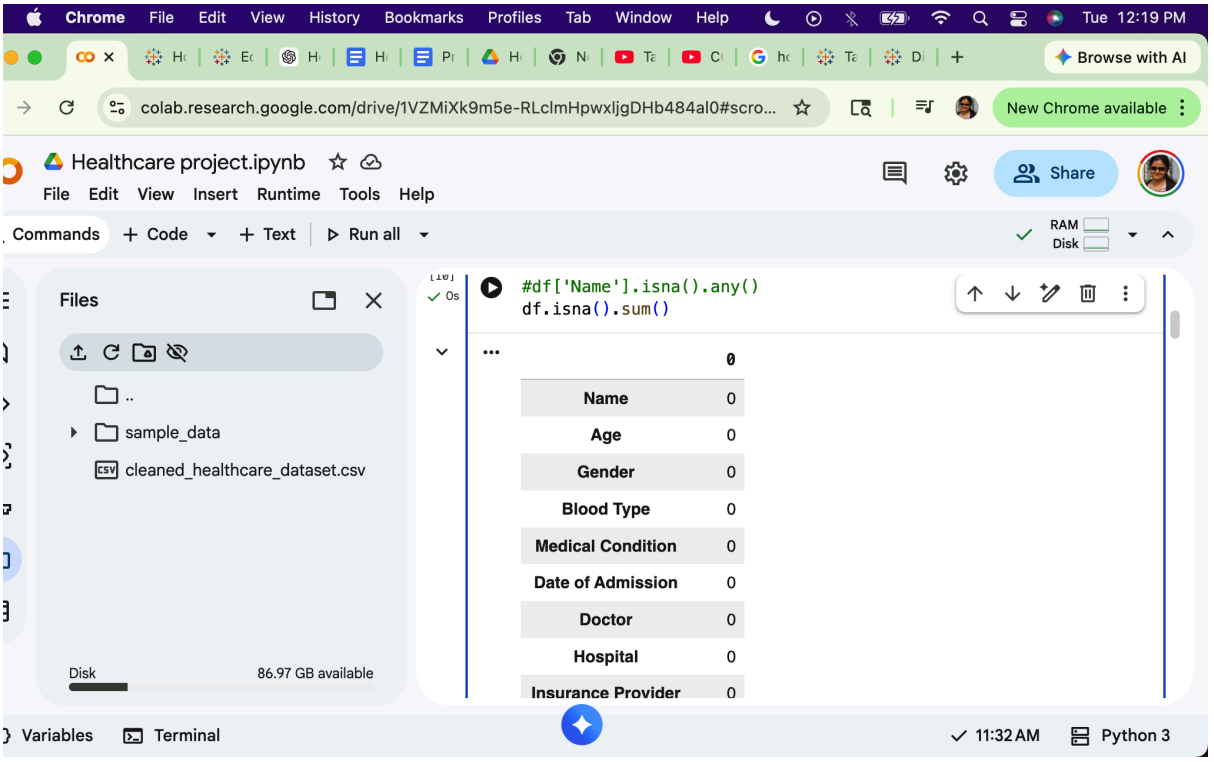
- **Name** – Patient name
- **Age** – Patient age
- **Gender** – Male/Female
- **Blood Type** – Blood group information
- **Medical Condition** – Disease diagnosed
- **Date of Admission & Discharge Date** – Hospital stay timeline
- **Hospital** – Hospital name
- **Doctor** – Attending doctor
- **Insurance Provider** – Insurance company
- **Billing Amount** – Cost of treatment
- **Room Number** – Assigned room
- **Admission Type** – Emergency, Urgent, etc.
- **Medication & Test Results** – Treatment details

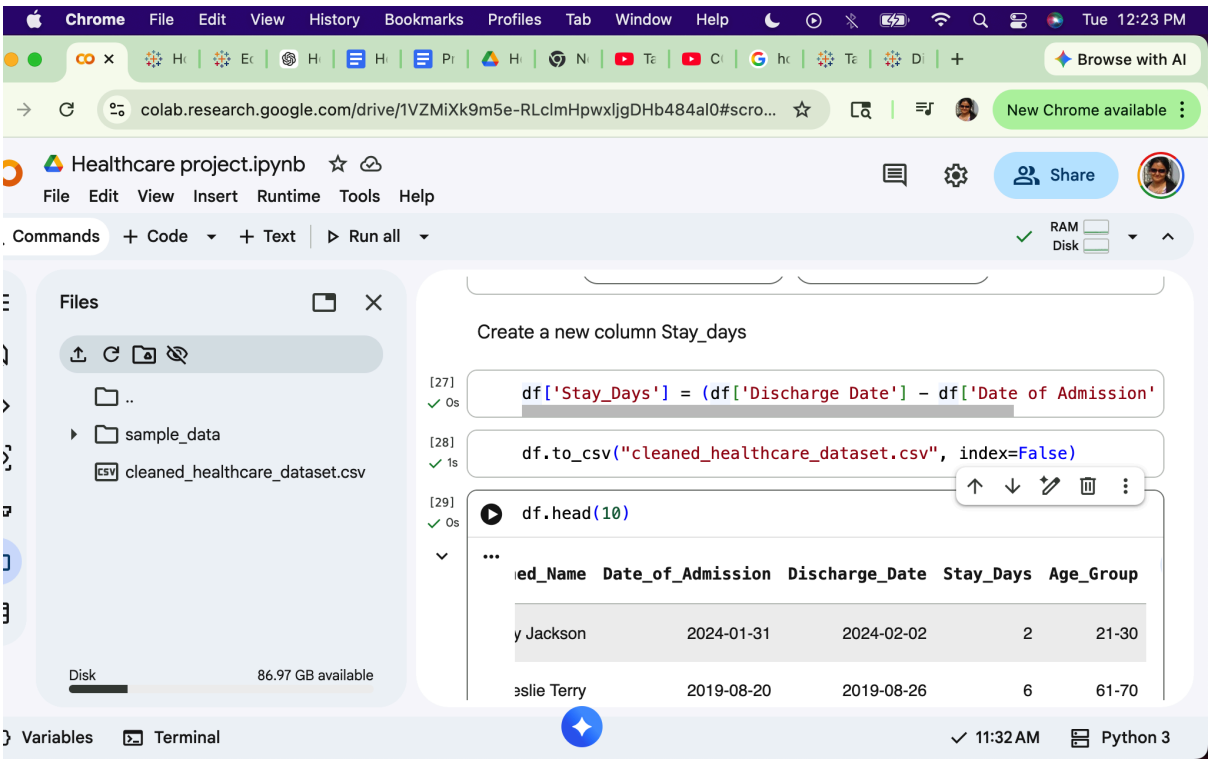
Tools Used

- Tableau: Data :Visualization and Dashboard creation
- Python (Pandas): Data Cleaning and Preprocessing

Data Cleaning and Preprocessing

The dataset was cleaned and prepared using Python with the following steps:

- Handled **missing values** using `.isna()` and appropriate treatments
- Converted date columns to **datetime format**
- Created a new column **Stay_Days**:
- 



Key Performance Indicators (KPIs)

The following KPIs were created in Tableau:

- **Total Patients** – Total number of patient records
- **Average Age** – Mean age of patients
- **Total Billing Amount** – Overall hospital revenue
- **Average Stay Duration** – Average number of days admitted
- **Top Disease** – Most common medical condition

Data Visualization

An interactive dashboard was created with the following visualizations:

1. Disease Distribution

- Bar chart showing number of patients per medical condition
- Helps identify the most common diseases

2. Age Group Distribution

- Categorizes patients into age groups (0–20, 21–40, 41–60, 60+)
- Shows demographic patterns

3. Billing by Hospital

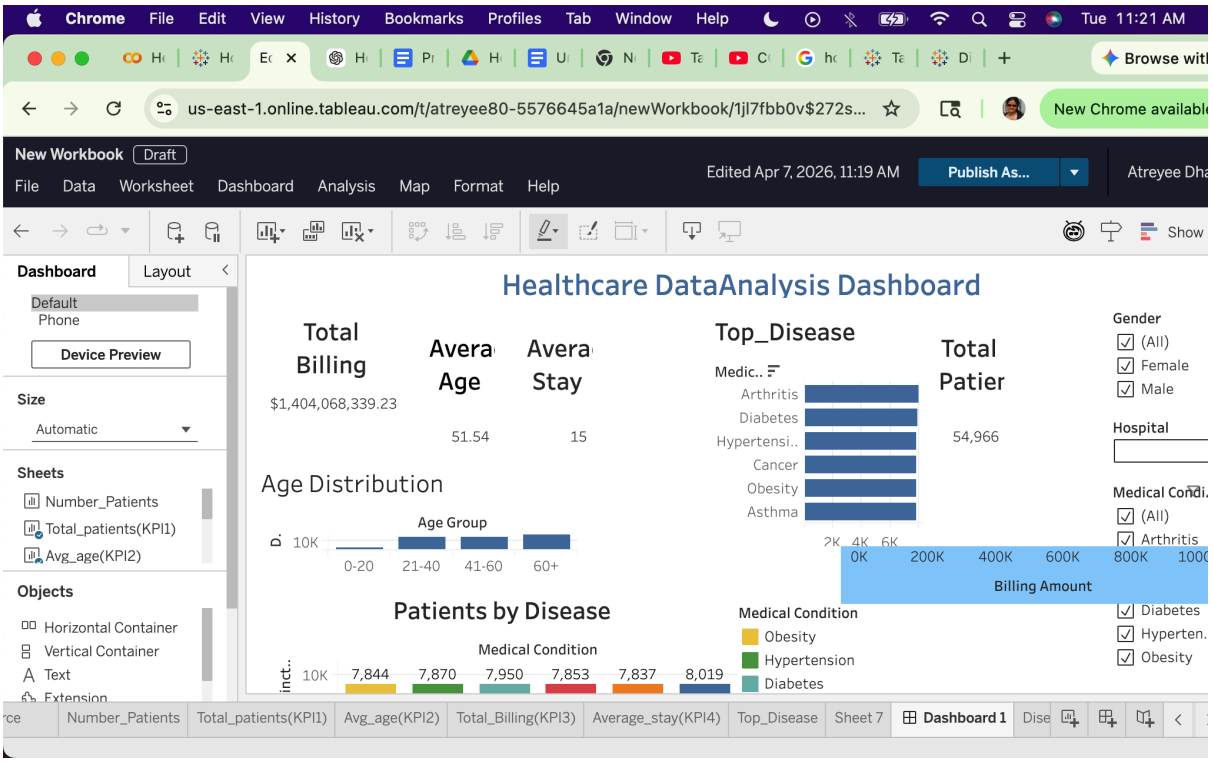
- Compares total billing across hospitals
- Identifies top-performing hospitals

Dashboard Features

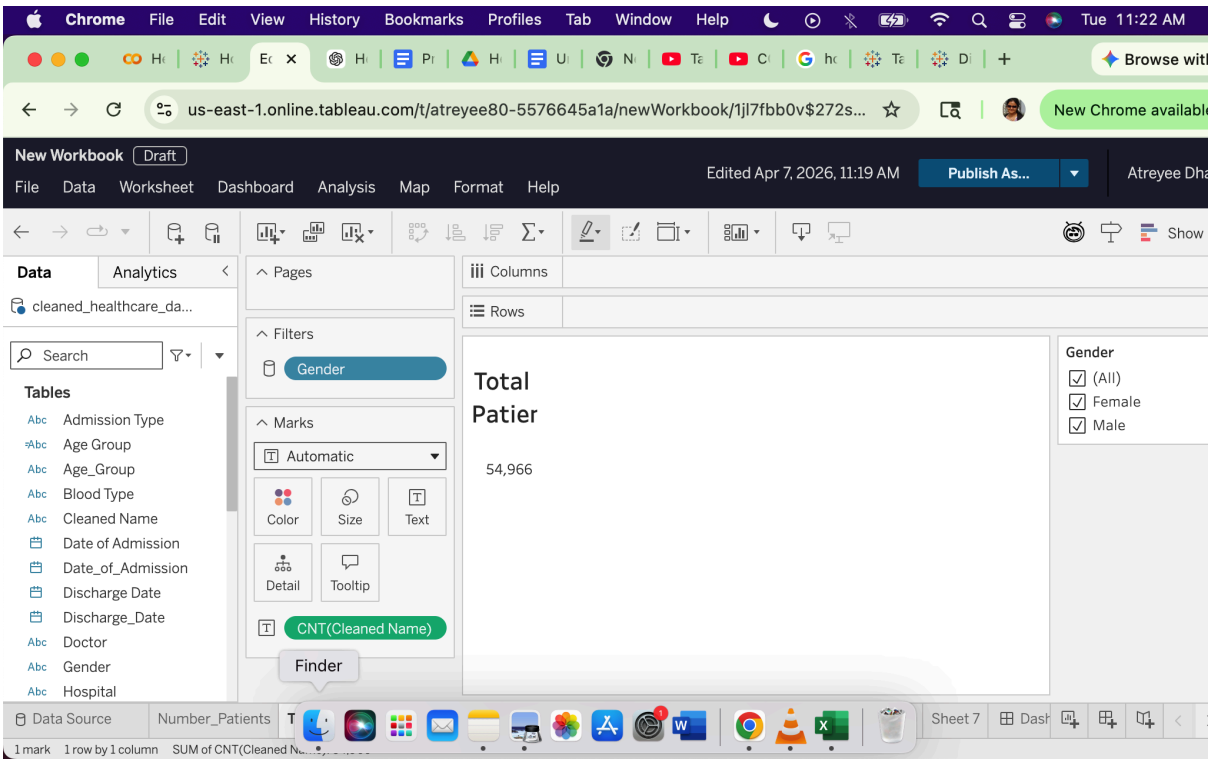
The Tableau dashboard includes:

- Interactive filters:
    - Gender
    - Hospital
    - Medical Condition
- Dynamic KPIs that update based on filters
- Clean and user-friendly layout for quick insights
-

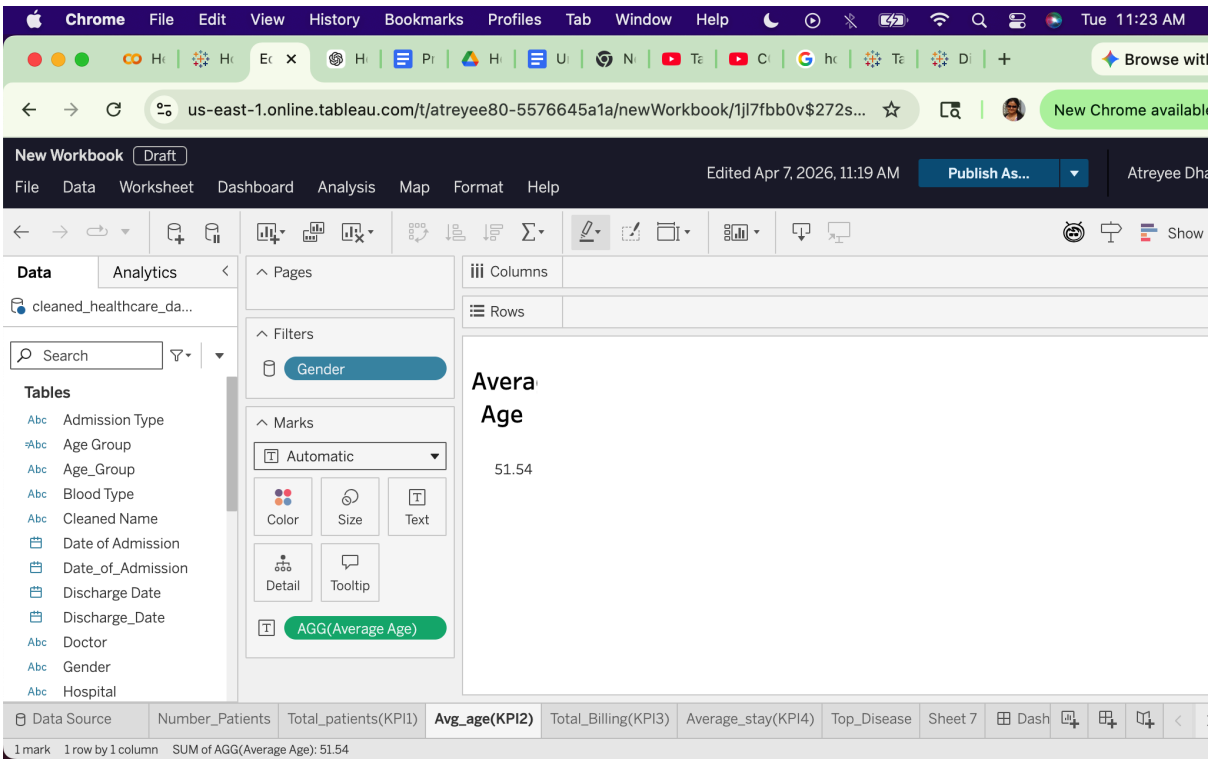
Dashboard Overview



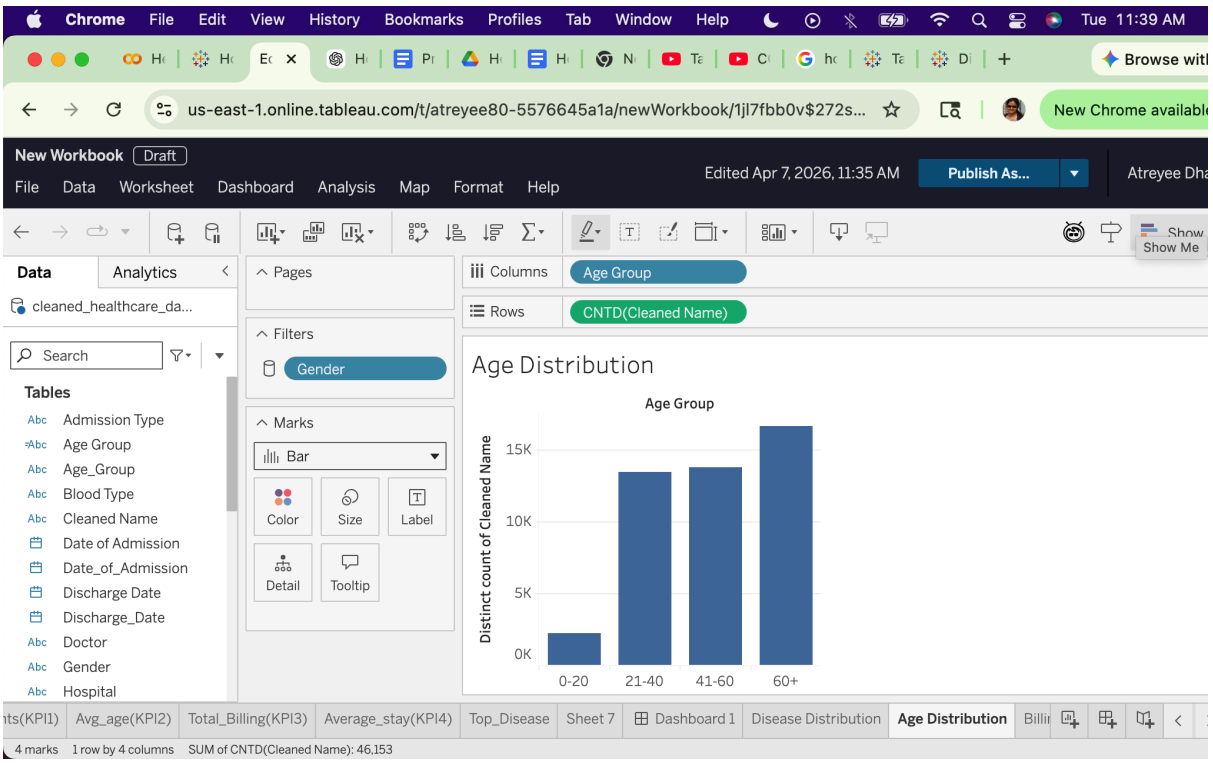
Total Patients



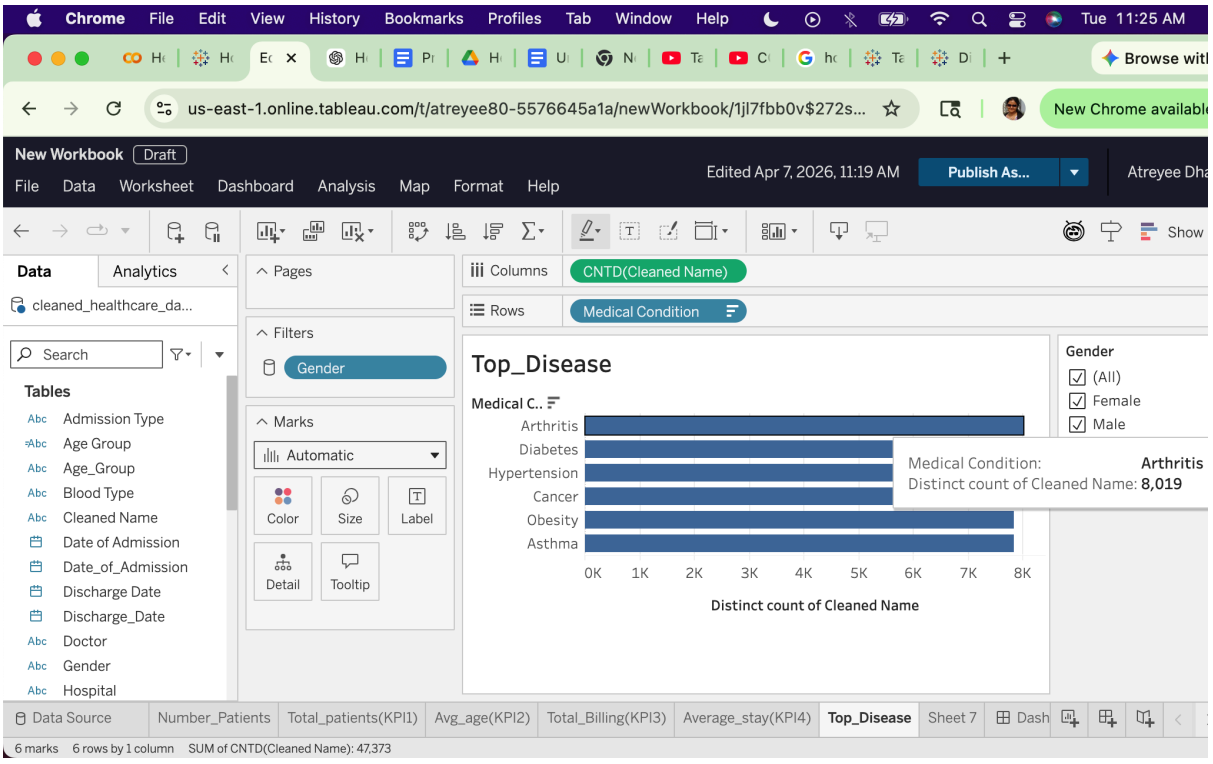
Average Age:



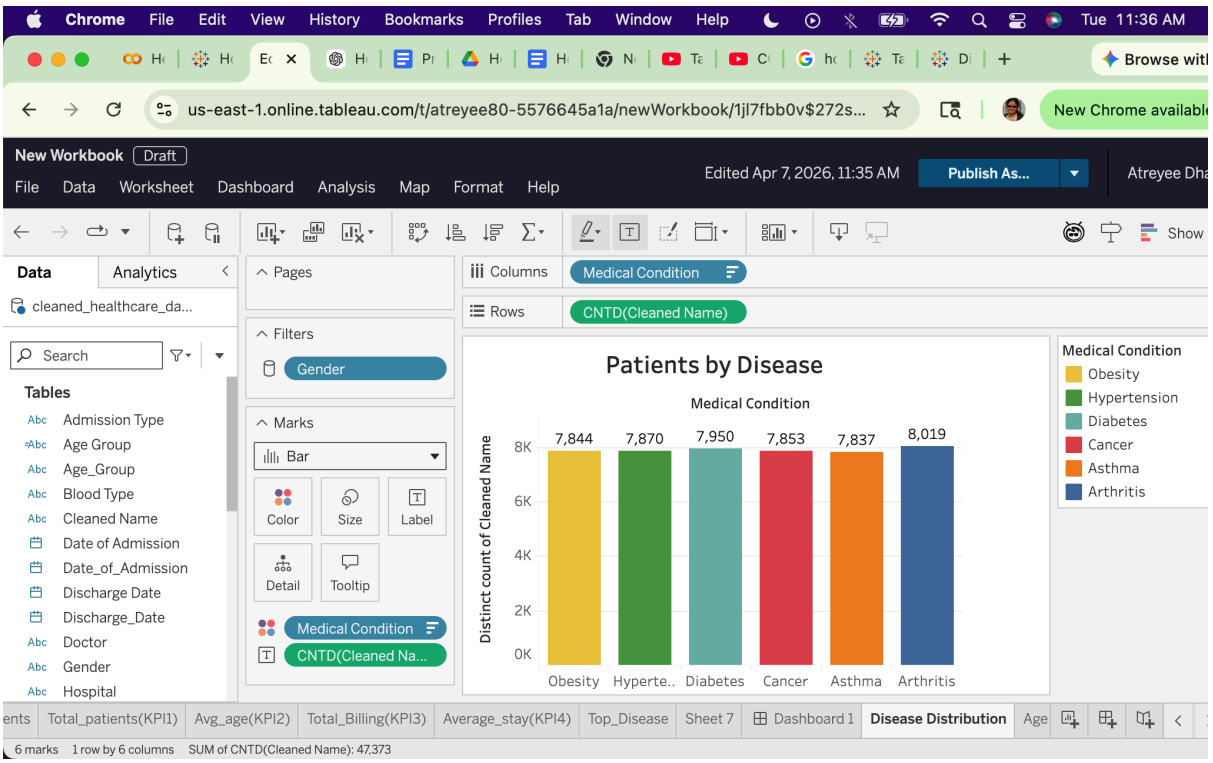
Age Distribution:



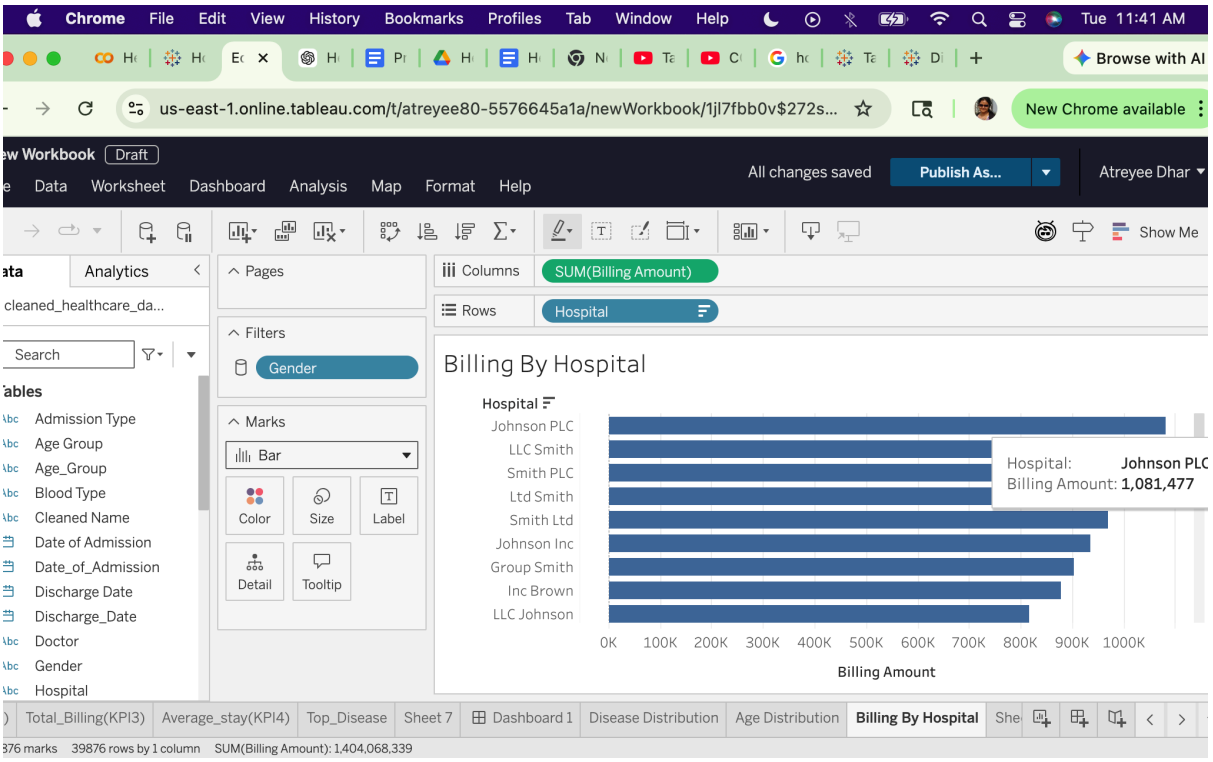
Top Disease:



Patients By Diseases:



Billing By Hospital:



Key Insights:

The most common disease is Arthritis 8019

Majority of the people fall in the age group of 50 or more
Total number of patients in hospitals is around 55000
Average hospital stay is 15days
Total Billing amount is $1404,068,339.23
Most common 3 diseases are Arthritis,Cancer and Asthma
Billing by hospital shows the top hospital amount is Johnson PLC $1081,477

Conclusion:

The dashboard helps in understanding better patients trends, disease prevalence, and hospital performance.

The interactive dashboard enables users to:

- Monitor key healthcare metrics
- Explore trends using filters
- Support data-driven decision-making

Future Improvements

- Add time-based analysis (monthly/yearly trends)
- Include predictive analysis using machine learning
- Expand dataset for more accurate insights
- Integrate real-time data sources